



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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COMMON HALF YEARLY EXAMINATION - 2024 - 25

Time Allowed : 3.00 Hours]

BIOLOGY

[Max. Marks : 70

Part - I (Bio-Botany) (Marks : 35)**Section - I****Note. (i) Answer All the questions.****8×1=8****(ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.**

- The first cell of male gametophyte in Angiosperm is
a) Microspore b) Megaspore c) Nucleus d) Primary endosperm nucleus
- Select the period for Mendel's Hybridization experiment
a) 1856- 1863 b) 1850-1870 c) 1857 - 1869 d) 1870-1877
- In which technique ----- ethidium Bromide is used
a) Southern Blotting Techniques b) Western Blotting Techniques
c) Polymerase chain Reaction d) Agrose gel Electrophoresis
- Assertion** : Turmeric fights Various kinds of Cancer
Reason : Curcumin is an antioxidant Present in Turmeric
a) Assertion is correct Reason is Wrong b) Assertion is Wrong Reason is Correct
c) Both are correct d) Both are Wrong
- is called the king of Bitter.
a) Nilavembu b) Basil c) Adathoda d) Turmeric
- People's movement for the Protection of environment in sirsi of Karnataka is
a) Chipko movement b) Amirtha Devi bishwas movements
c) Appiko movement d) None of the above
- Which one of the following Crop varieties correct matches with its resistance to disease?

Variety	-	Resistance of Disease
a) Pusa Komal	-	Bacterial Blight
b) Pusa Sadabahar	-	White Rust
c) Pusa Shubhra	-	Chilli Mosaic Virus
d) Brassica	-	Pusa Swarnim
- The unit for measuring ozone thickness.
a) Joule b) Kilos c) Dobson d) Watt

SECTION - II**II. Answer any four questions.****4×2=8**

- Draw and Label the Structure of a mature embryo sac of angiosperms.
- Differentiate missense and nonsense mutation.
- Write two advantages of Artificial Seeds.
- Give four examples of Plants cultivated in commercial Agroforestry.
- What is Biopesticides.
- Sandy soil is not suitable for cultivation Explain. Why?

SECTION - III**III. Answer any three questions. Question No. 18 is compulsory.****3×3=9**

- What is Gene mapping? write its uses?
- Write a note on Biolistics.
- Give an account of various types of parasitism with example.
- Write the Steps involved in Hybridization.
- Write the Advantages of Conventional methods of vegetative reproduction in Plants.

SECTION - IV**IV. Answer all the following:****2×5=10**

- (a) Enumerate the Characteristic feature of Entomophilous Flowers.
(OR)
(b) Explain the Technique involved in Plant tissue Culture.
- (a) What are the king and Queen of Spices? Explain about them and Their Uses.
(OR)
(b) Write the Significance of Plant Succession.



PART - II (BIO-ZOOLOGY) (Max. Marks: 35) SECTION - I

Note: (i) Answer all the questions

(ii) Choose the most appropriate answer from the given four Alternatives and write the option code and corresponding answer.

- Identify the correct statements from the following
 - Chlamydia is a viral disease.
 - Gonorrhoea is caused by a spirochaete bacterium, *Treponema palladium*.
 - The incubation period for syphilis is 2 to 14 days in males and 7 to 21 days in females.
 - Both syphilis and gonorrhoea are easily cured with antibiotics.
- The first codon to be deciphered was ----- which codes for -----
 - AAA, proline
 - GGG, alanine
 - UUU, Phenylalanine
 - TTT, arginine
- The age of fossils can be determined by
 - electron microscope
 - weighing the fossils
 - carbon dating
 - analysis of bones
- Which of the following statements support the view that elaborate sexual reproductive process develops much later in the organic evolution?
 - Lower groups of organisms have simpler body design
 - Asexual reproduction is common in lower groups
 - Asexual reproduction is common in higher groups of organisms
 - The high incidence of sexual reproduction is in angiosperms and vertebrates.
 - i, ii and iii
 - i, iii and iv
 - i, ii, and iv
 - ii, iii and iv
- Transgenic animals are those which have
 - Foreign DNA in some of their cells
 - Foreign DNA in all their cells
 - Foreign RNA in some of their cells
 - Foreign RNA in all their cells
- Competition between species leads to
 - Extinction
 - Mutation
 - Amensalism
 - Symbiosis
- Assertion:** The Environmental conditions of the tropics are favourable for speciation and diversification of organisms
Reason: The climate seasons, temperature, humidity and photoperiod are more or less stable and congenial.
 - Both Assertion and Reason are true and Reason explains Assertion correctly.
 - Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
 - Assertion is true, but Reason is false.
 - Both Assertion and Reason are false.
- The active form of *Entamoeba histolytica* feeds upon
 - Erythrocytes, mucosa and submucosa of colon
 - Mucosa and submucosa colon only
 - Food in intestine
 - Blood only

SECTION - II

II. Answer the following any four question.

- Differentiate J shaped and S- shaped curve.
- What are stem cells? Explain the role in the field of medicines.
- Mention the symptoms of down's syndrome
- Draw a neat labeled diagram of human ovum.
- What is anaphylaxis?
- Red data book-what do you know about it?

SECTION - III

III. Answer any 3 of the following . Question no 19 is compulsory.

- Mention the importance of the position of the testes in humans.
- Discuss the role of an individual to reduce environmental pollution.
- What are lactic acid bacteria? Give examples
- What are called molecular clocks? Which molecules are used to study evolution?
- The genes for Haemophilia are located on the chromosomes of humans. It is normally impossible for a hemophiliac father to pass the gene to his son. Why?

SECTION - IV

IV. Answer in Detail.

- Write a brief note on the causative agent, symptoms, diagnostic and preventive measures of cervical cancer
 (OR)
 Name and explain any one of the lymphoid organs present in humans.
- List out the various causes of biodiversity losses.
 (OR)
 Explain the formation of nucleosome.